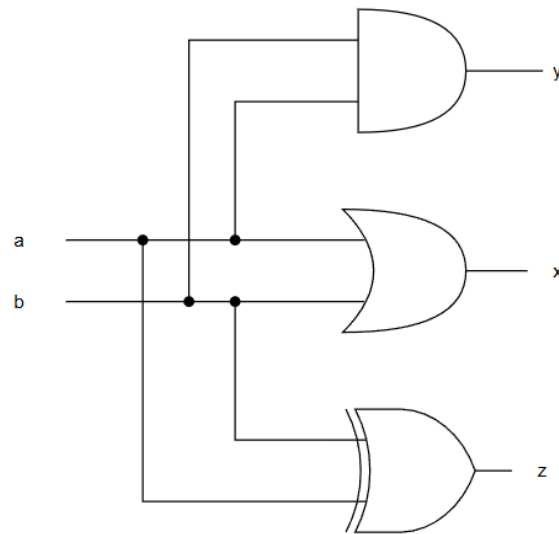


Task 1 & 2

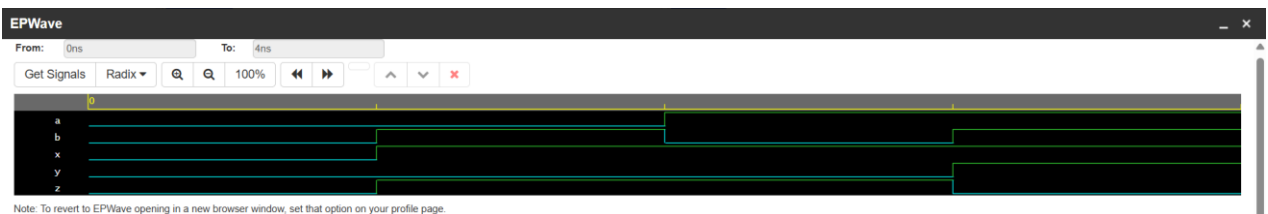
<https://edaplayground.com/x/V47b>



Gate Level Diagram

```
# KERNEL: SLP loading done - time: 0.0 [s].
# KERNEL: Warning: You are using the Riviera-PRO EDU Edition. The performance of simulation is reduced.
# KERNEL: Warning: Contact Aldec for available upgrade options - sales@aldec.com.
# KERNEL: SLP simulation initialization done - time: 0.0 [s].
# KERNEL: Kernel process initialization done.
# Allocation: Simulator allocated 4545 kB (elbread=427 elab2=4111 kernel=6 sdf=0)
# KERNEL: ASDB file was created in location /home/runner/dataset.asdb
# KERNEL: [PASS] ab=00 AND: value=0
# KERNEL: [PASS] ab=00 XOR: value=0
# KERNEL: [PASS] ab=00 OR: value=0
# KERNEL: [PASS] ab=01 AND: value=0
# KERNEL: [PASS] ab=01 XOR: value=1
# KERNEL: [PASS] ab=01 OR: value=1
# KERNEL: [PASS] ab=10 AND: value=0
# KERNEL: [PASS] ab=10 XOR: value=1
# KERNEL: [PASS] ab=10 OR: value=1
# KERNEL: [PASS] ab=11 AND: value=1
# KERNEL: [PASS] ab=11 XOR: value=0
# KERNEL: [PASS] ab=11 OR: value=1
# RUNTIME: Info: RUNTIME_0068 testbench.sv (35): $finish called.
# KERNEL: Time: 4 ns, Iteration: 0, Instance: /tb, Process: @INITIAL#24_00.
# KERNEL: stopped at time: 4 ns
# VSIM: Simulation has finished. There are no more test vectors to simulate.
# VSIM: Simulation has finished.
Done
```

Terminal Output Screenshot



Waveform Screenshot

